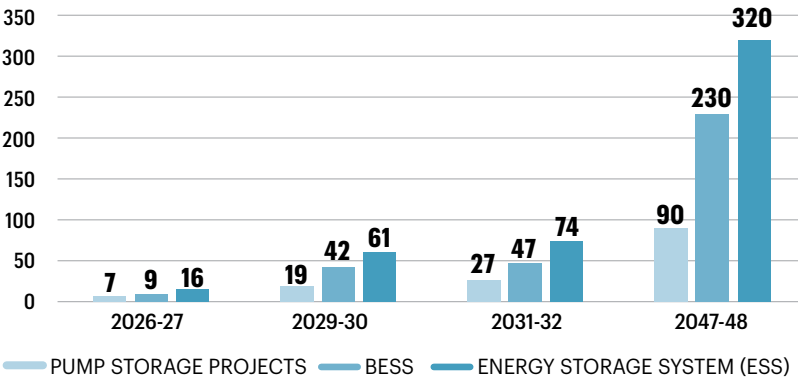


TAKING STOCK

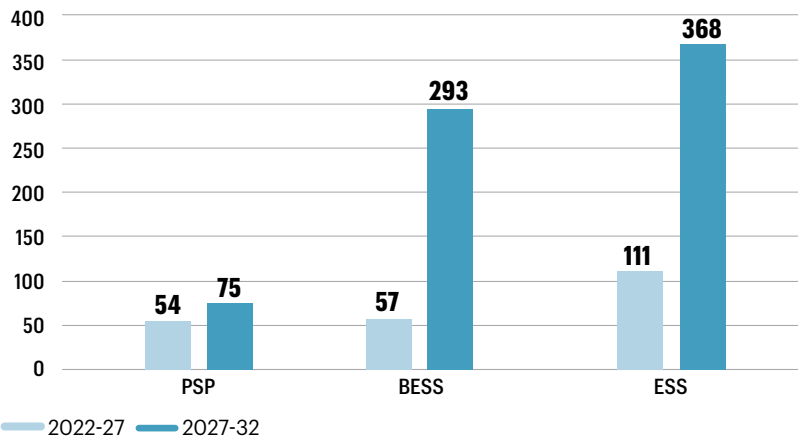
- India's energy storage sector is projected to grow 5X
- ESS will require an investment of ₹368 crore by 2032

ESTIMATED ENERGY STORAGE CAPACITY



NOTE FIGURES IN GW

ESTIMATED INVESTMENTS



NOTE FIGURES IN ₹CR SOURCE NATIONAL FRAMEWORK FOR PROMOTING ENERGY STORAGE SYSTEMS

**₹18,000
CRORE**

**OUTLAY FOR PLI SCHEME
TO SUPPORT 50GWH
ADVANCED CHEMISTRY
CELL PRODUCTION**

2023 to ₹3.72 lakh/MW/month in a 250MW sale in March 2024 by the Gujarat Urja Vikas Nigam Ltd.

They fell even further by 41.3% to ₹2.19 lakh/MW/month in a 500MW auction in November 2024 by Rajasthan, supported by VGF and expected declines in bat-

tery costs, says Charith Konda, an energy specialist at the Institute for Energy Economics and Financial Analysis.

In fact, a graph of tendered capacity of standalone BESS projects exhibits a hockey stick trajectory—a long period of slow or flat growth followed by a sharp increase on the back of factors such as VGF. Solar energy projects co-located with storage are, meanwhile, growing at a steady pace.

“If you look at the capacities, 7GWh of BESS capacity was cancelled out of around 30GWh between 2018 and 2023, and this reduced to zero in 2025 (till now). This tells you what VGF can do to a sector,” says Debmalaya Sen, President of the India Energy Storage Alliance (IESA).

Costs are no longer just about cells. They include integration, software, safety systems and duties. While raw material price volatility in lithium, cobalt and nickel may cause short-term bumps, the bigger shift in India will come from economies of scale, localisation and innovative financing, say experts.

“Over the medium term, as PLI-backed giga factories come online and chemistries like sodium-ion and solid-state mature, we expect storage costs to decline another 20–30% by 2030. That will make storage not just viable, but inevitable, for India’s power systems,” says Vineet Mittal, Chairman of Avaada Group, a renewable energy company. PLI is short for production-linked incentives, or benefits offered by the government to boost domestic manufacturing and global competitiveness in select industries. Paced by government incentives and falling prices of battery packs, turnkey BESS costs tumbled 40% in 2024.